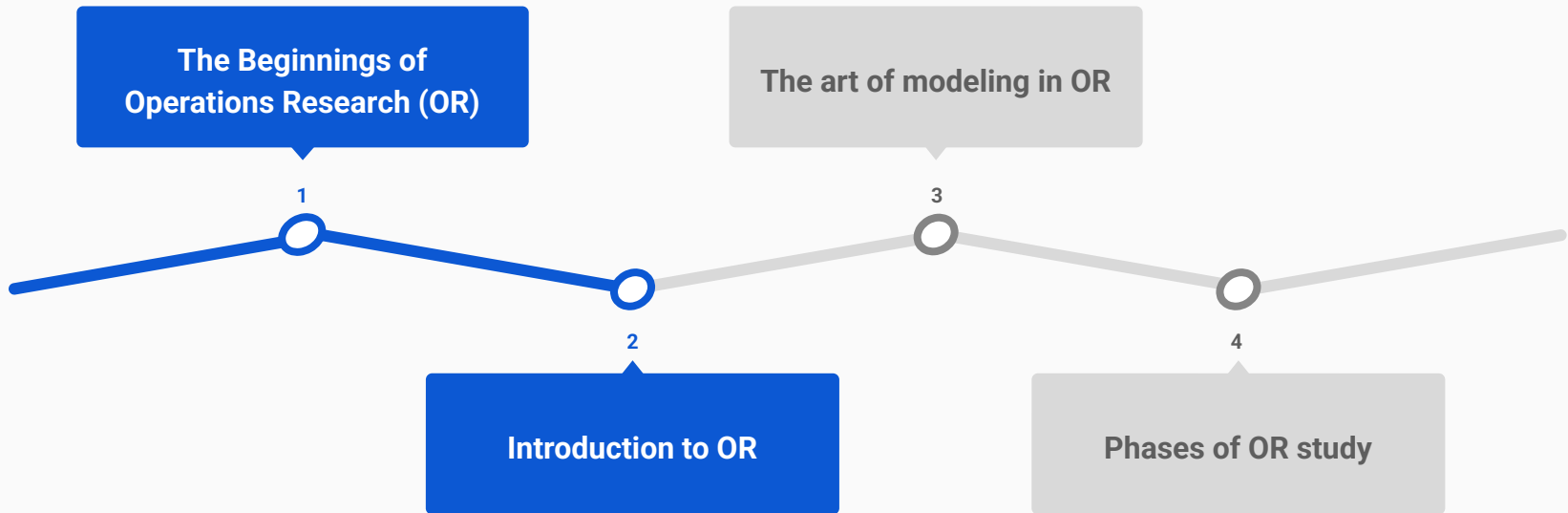


The art of solving real-world problems: Operations Research (OR)

Mohammed Brahim



Agenda



The Beginnings of Operations Research (OR)

Origine of Operations Research (OR)

Who were the first Operations Research teams?



***The scientists became the first operations research (OR) teams
During World War II***

Origine of Operations Research (OR)

- Large number of scientists are called to apply a scientific approach to address strategic and tactical problems
- Results
 - Scarce resources allocation to military operations
 - Deployment of radar
 - Convoy management
 - ...

The origins of operations research (OR) is the a time of war, when highly intelligent scientists came together to tackle *crucial problems*.

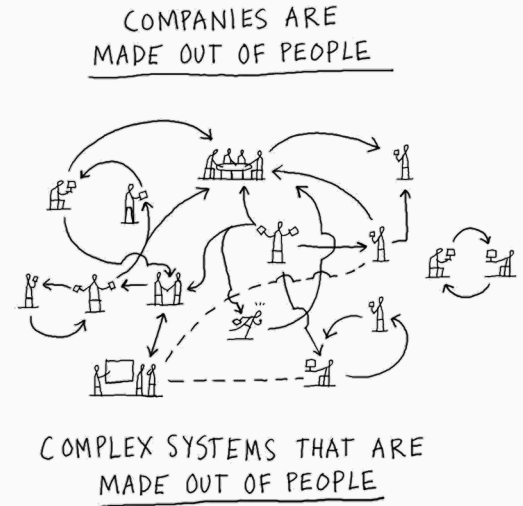
OR from War to Business and Research



- OR Teams during War spur Adoption and Advancements in 1950s
 - Business consultants introduced OR to various organizations in the early 1950s
 - Scientists motivated to pursue OR research, leading to significant advancements in the field
- Simplex Method revolutionizes Linear Programming in 1947
- OR Advances in 1950s: Linear Programming, Dynamic Programming, Queuing, Inventory Theory

OR and organizational Growth Challenges

- Growth of organizations
- Emergence of components with separate goals
- Conflict between components
- Difficulty in resource allocation



**Emergence of Operations Research (OR) to
solve these problems**

OR and computer revolution

- The computer revolution was a major boost to OR's development
- The continued acceleration of computer power has contributed to the growth of OR



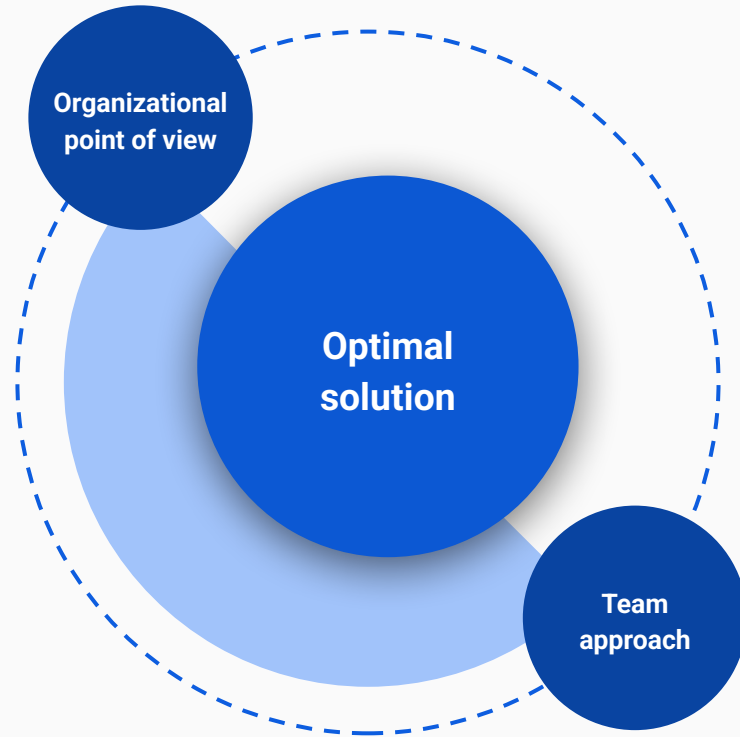
Introduction to Operations Research (OR)

Operations Research

- **Operations** : problems related to operations within organizations
- **Research** : scientific method to investigate the problem
 - Observation and formulation
 - Building a mathematical model
 - Testing the model.

Nature of Operations Research (OR)

- OR adopts an **organizational point of view** and considers the best for the organization as a whole
- OR searches for the **best possible solution (optimal)** not only an acceptable solution
- OR requires a **team approach**, as no single individual can have expertise in all aspects of OR



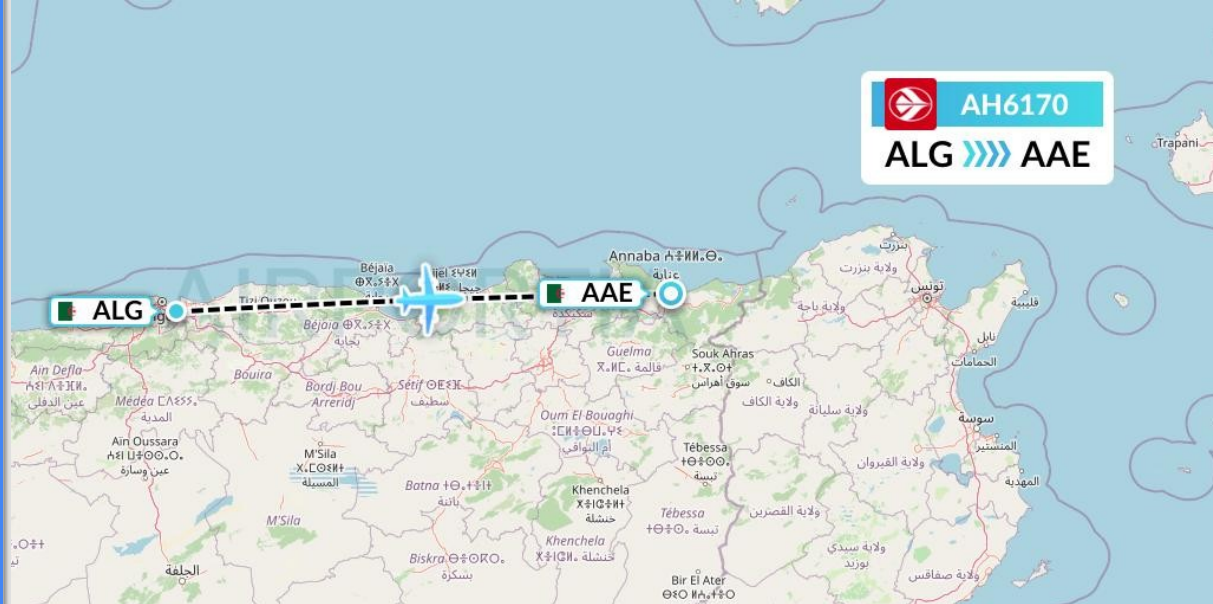
Impact of Operations Research (OR)



Organization	Use case	Annual Savings
Samsung Electronics	Reduce manufacturing time and the amount of stocked raw material	\$200 million more revenue
Hewlett-Packard (HP)	Product portfolio management	\$180 million
Netherlands Railways	Optimize operation of a railway network	\$105 million
Waste Management	Develop a route management system for trash collection and disposal	\$100 million
Syngenta	Increase the productivity of crops	\$57 million

Example: Tickets purchasing problem

- A businessperson has a 5-week commitment traveling between Annaba (AAE) and Algiers (ALG)
- Weekly departure from Annaba (AAE) is on Mondays and returns on Wednesdays
- A regular round trip ticket costs \$400
- A 20% discount is granted if the roundtrip dates span a weekend.
- A one-way ticket costs 75% of the regular roundtrip ticket price.



Determine the best way to purchase tickets for the 5-week period

Tickets purchasing problem

Alternatives

What are the decision alternatives ?

Constraints

What are the restrictions for making the decision?

Objective

What is an appropriate objective criterion for evaluating the alternatives ?

What are the decision alternatives ?

- a. Buy five regular AAE-ALG-AAE for departure on Monday and return on Wednesday of the same week.
- b. Buy one AAE-ALG, four ALG-AAE-ALG that span weekends, and one ALG-AAE.
- c. Buy one AAE-ALG-AAE to cover Monday of the first week and Wednesday of the last week and four ALG-AAE-ALG to cover the remaining legs. All tickets in this alternative span at least one weekend.

Tickets purchasing problem

What are the restrictions for making the decision?

The businessperson must leave AAE on Monday and return on Wednesday of the same week.

Objective criterion

The cheapest ticket price.

Alternatives costs

- a. Alternative a : Cost: \$2000 ($5 * \400)
- b. Alternative b : Cost: \$1880 ($0.75 * \$400 + 4 * 0.8 * \$400 + 0.75 * \400)
- c. Alternative c : Cost: \$1600 ($5 * 0.8 * \400)**

The best alternative is the one with the smallest cost

Challenge : what are the decision alternatives ?

- How many alternatives exist?
- Is there a cheaper alternative to the options offered ?
- What is the cost of this alternative?

Example: Garden problem

- A homeowner is starting a backyard vegetable garden
- The garden must be rectangular and fenced to keep critters out
- The owner has enough material to build a fence of length $L = 100$ ft



**Fence the largest possible
rectangular area**

Garden problem

Alternatives

What are the decision alternatives ?

Constraints

What are the restrictions for making the decision?

Objective

What is an appropriate objective criterion for evaluating the alternatives ?

What are the restrictions for making the decision?

- We control the **width (w)** and **height (h)** of the rectangle
- Width, height are continuous variables between 0 and L.
- Perimeter of the garden ≤ 100

Objective criterion

Maximize the area of the garden

Alternatives costs

Impossible to test all the possibilities because we have continuous variables

Garden problem

Maximize $z = wh$

subject to

$$2(w + h) = 100$$

$$w, h \geq 0$$

Operations Research

problem formulation

Decision variables: variables that we can control to optimize the objective function

Feasible solution : solution satisfies all constraints

Optimal solution : It yields the best (maximum or minimum) value of the objective function

Maximize or minimize
Objective Function
Subject to
Constraints

The art of modeling in Operations Research (OR)

Example: Wozac production process

- Eli Daisy produces Wozac by heating a chemical mixture in a pressurized container
- The production process results in different amounts of Wozac produced each time
- The amount produced is called the process yield, measured in pounds



**Understand the factors that
influence the yield.**

- **Model-building** process should be used
- **Decision variables:** factors influencing Wozac production yield:
 - Container volume (V), Container pressure (P), Container temperature (T)
 - A, B, and C be percentage of mixture made up of chemicals A, B, and C
- **Model:** descriptive model that describes behavior of actual yield
 - Regression methods used to determine influence
 - Find the model parameters based on the data

$$\begin{aligned} yield = 300 &+ 0.8V + 0.01P + 0.06T + 0.001T * P - 0.01T^2 - 0.001P^2 \\ &+ 11.7A + 9.4B + 16.4C + 19A * B + 11.4A * C - 9.6B * C \end{aligned}$$

Constraints in Wozac Production Process

What are the restrictions for making the decision on Wozac Production Process ?

- Volume (V) between **1 and 5 liters**
- Pressure between **200 and 400 milliliters**
- Temperature between **100 and 200 degrees Celsius**
- Mixture must be made of A, B, and C
- Maximum half of mixture can be product A for proper performance of the drug

Wozac Production Process

$$\begin{aligned}\text{Maximize } z = & 300 + .8V + .01P + .06T + .001T*P - .01T^2 - .001P^2 \\ & + 11.7A + 9.4B + 16.4C + 19A*B + 11.4A*C - 9.6B*C\end{aligned}$$

Subject to (s.t.)

$$V \leq 5$$

$$V \geq 1$$

$$P \leq 400$$

$$P \geq 200$$

$$T \leq 200$$

$$T \geq 100$$

$$A \geq 0$$

$$B \geq 0$$

$$C \geq 0$$

$$A + B + C = 1$$

$$A \leq 5$$

Wozac Production Process

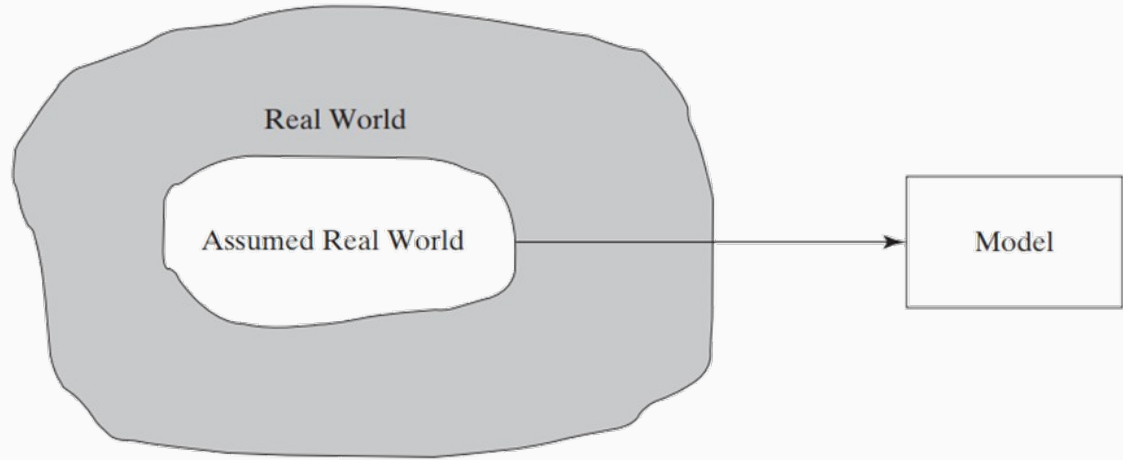
- Any values of decision variables that meet constraints is in **feasible region**
- **Optimal solution** is a point in feasible region that optimizes objective function
- Use an OR method to determine the optimal solution for the model



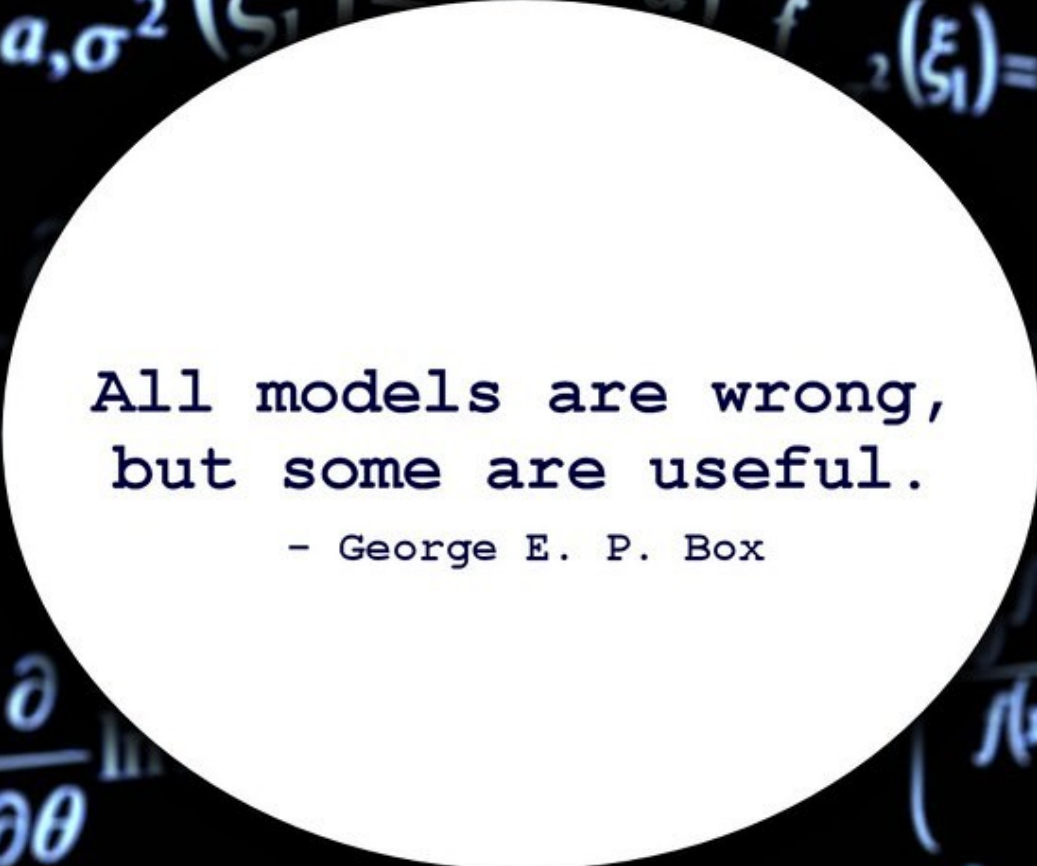
- **Optimal solution:**
 - 5-liter container, 200 milliliters pressure, 100 degrees Celsius temperature, 29% A and 71% C
 - Maximum yield of **183.38 pounds** can be obtained with the **optimal solution**

Modeling in OR

- Model expresses the behavior in an easily understandable manner
- Dominant variables are concentrated
- Mathematical functions representing the behavior of the assumed real world



Model is a simplified representation of reality, not reality itself



All models are wrong,
but some are useful.

- George E. P. Box

Solving OR model

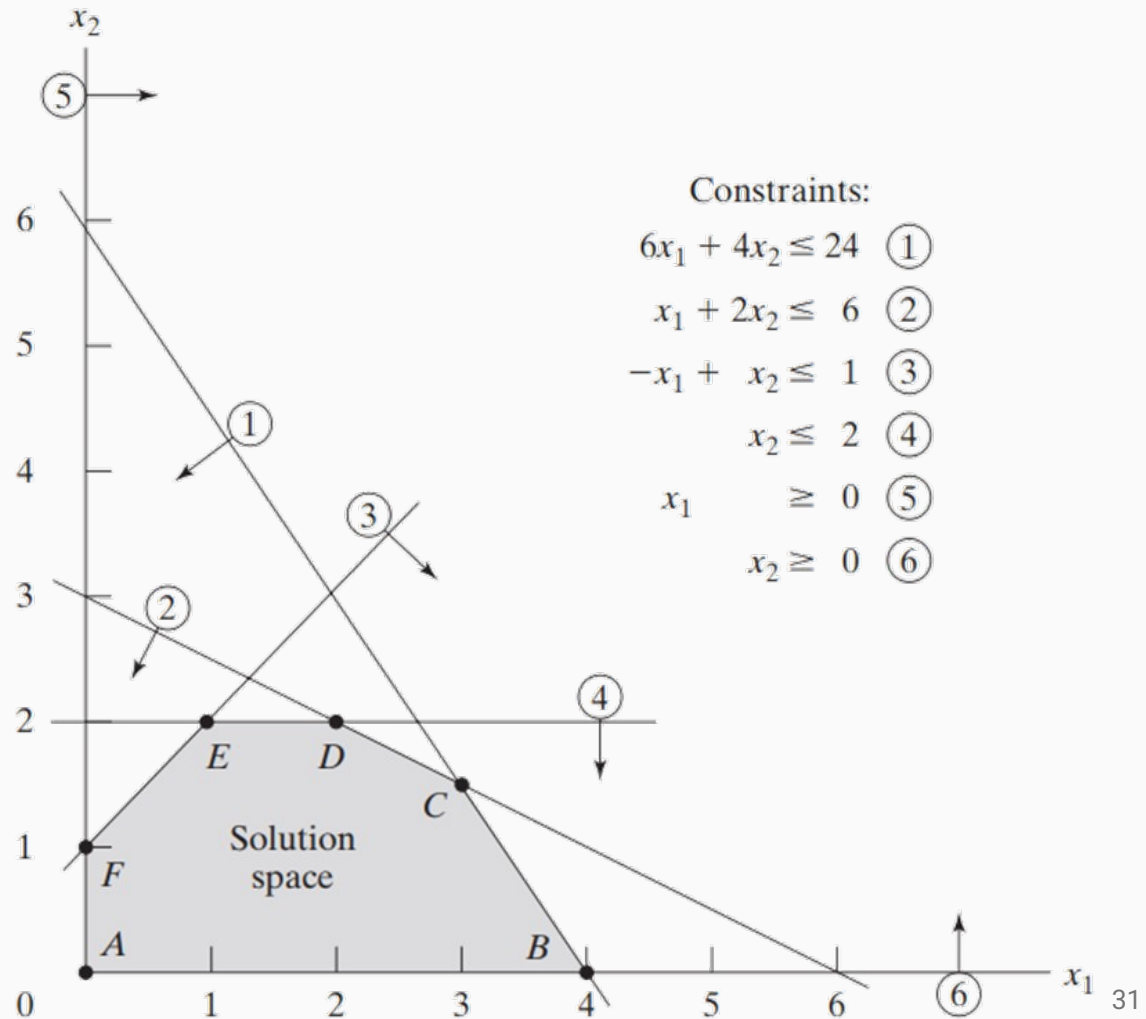
- OR Technique

- Linear Programming is the most prominent one (next week)

- Other OR Techniques

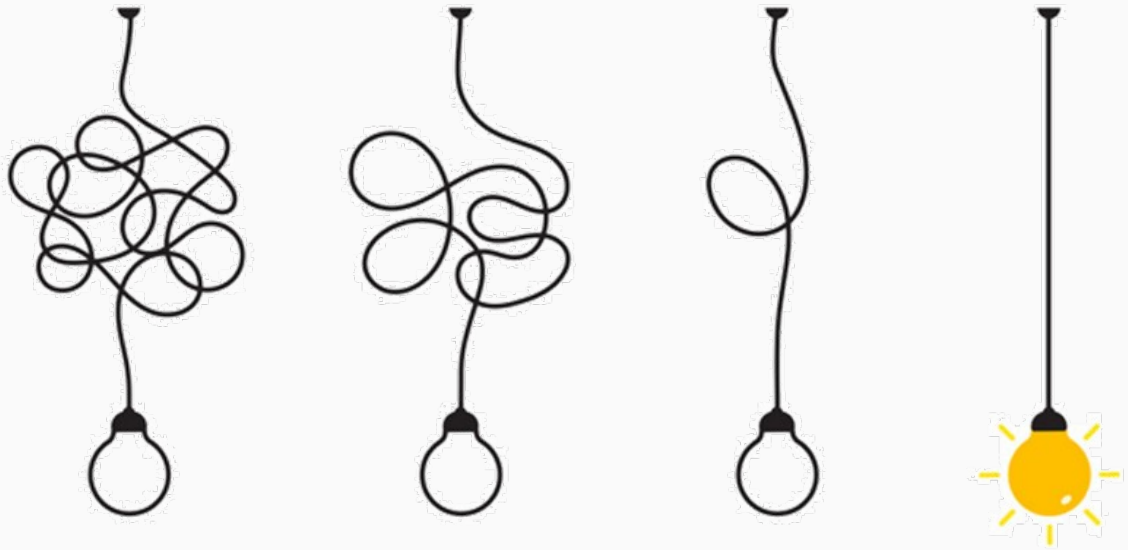
- Integer Programming
- Dynamic Programming
- Network Programming
- Nonlinear Programming

- ...



Modeling is more than Just Mathematics

- OR models rely on mathematical analysis
- Start Simple, Then Progress to Complex Models
- Human elements play a crucial role in decision making
- Understanding psychology is important



**Avoid Mathematical Bias,
Prioritize Understanding & Simple Solutions**

Example: UPS- Optimize delivery itineraries

Problem:

- UPS introduced ORION software to provide tailored delivery itineraries
- The software proposed shorter routes, leading to potential savings
- **Drivers didn't like the idea of a machine "beating" them**





Problem:

- UPS introduced ORION software to provide tailored delivery itineraries
- The software proposed shorter routes, leading to potential savings
- **Drivers didn't like the idea of a machine "beating" them**

Solution:

- ORION developers added a button **"Beat the Computer"**
- Drivers took the challenge, some beat computer-suggested routes
- Perception of ORION changed from a **threat** to a **complement to drivers' experience**

Example: Solving the Airport Baggage Delay



Problem:

- Long baggage wait complained about by Houston Intercontinental Airport travelers
- **Increasing the number of baggage handlers did not alleviate the problem**

Solving the Airport Baggage Delay



Problem:

- Long baggage wait complained about by Houston Intercontinental Airport travelers
- **Increasing the number of baggage handlers did not alleviate the problem**

Solution:

- **Arrival gates were moved farther away from baggage claim**
- **Extra walking allowed ample time for luggage to be delivered to the carousel**
- **Complaints disappeared**

Example: Elevator Wait Time Issues

Problem:

- Complaints of slow elevator service in a large office building
- **Waiting-line problem requiring mathematical queuing analysis**



Elevator Wait Time Issues

Problem:

- Complaints of slow elevator service in a large office building
- **Waiting-line problem requiring mathematical queuing analysis**



Solution proposed with psychologist:

- **Installation of full-length mirrors at the entrance to the elevators**
- **People were kept occupied watching themselves and complaints disappeared**

Example: Solving the Long Wait for Truck Transport

Problem:

- A number of departments share three trucks to transport material
- Departments complained of long wait and demanded adding a fourth truck
- **The problem was perceived as a need for an additional truck**



Solving the Long Wait for Truck Transport

Problem:

- A number of departments share three trucks to transport material
- Departments complained of long wait and demanded adding a fourth truck
- **The problem was perceived as a need for an additional truck**



Solution:

- Low truck utilization due to obscure parking and lack of visibility for supervisors
- Two-way radio communication was installed to solve the problem
- **Moving trucks to a more visible location to increase visibility and encourage usage**

Lessons Learned from these examples

1. Consider diverse ideas, seek social **science/psychology expertise**
2. **Avoid bias to use mathematical tools**, use simplest technique to analyze data
3. Adopt a bird's eye view to **uncover nontechnical reasons for the problem**
4. **Consider human behavior in solutions**, root in people not just technology

Phases of an Operations Research (OR) Study

Phases of an OR Study (1)

Define the problem

Observe the system

Build the [math] model

Solve the model

Validate the model

Present the results

Implement the final solution

Monitor the system

Phases of an OR Study (2)

Define the problem

- Define the problem
 - Objectives
 - Decision alternatives
 - Constraints
- Determine the study scope
 - Organization parts

Observe the system

- Collect data
- Data tells story about the problem.
- Determine the variables affecting problem

Build the [math] model

- Translate problem to a math model
- Model may fit standard math models (e.g. linear programming)

Phases of an OR Study (3)

Solve the model

- Use well-defined methods if possible
- Heuristics or simulation if not
- Sensitivity analysis (to be covered)

Validate the model

- Validate the model's output with historical data or simulations
- Ensure the solution is intuitively acceptable

Present the results

- Present model and recommendation to decision maker
- Discuss feasibility
- Revisit earlier phases if recommendations are not approved

Phases of an OR Study (4)

Implement the final solution

- Assistance in implementing the recommendations
- Translating the results into clear and understandable operating instructions

Monitor the system

Monitor the system and update it regularly to meet organization's objectives

Example: CITGO Petroleum



- CITGO is an oil-refining and marketing company
- Klingman et al. (1987) applied management-science techniques to CITGO Petroleum
- The work saved the company an estimated \$70 million per year
- CITGO was purchased by Southland Corporation (the owners of 7-Eleven stores)

Phases of an OR Study (1)

Define the problem

- The goal : Minimize the cost of operating CITGO's refineries

Observe the system

- Observe Lake Charles Refinery
- Estimate key relationships (cost of producing each product, energy needed to produce each product, yield, etc.)
- Install Accurate Data Collection Systems

Build the [math] model

- Develop Linear programming (LP) model to optimize operations

Phases of an OR Study (2)

Solve the model

- Solve the Linear programming problem using a software package

Validate the model

- Validate model with actual inputs and outputs from the refinery

Present the results

- Show Benefits of \$50M+ Annually to Decision Makers

Phases of an OR Study (2)

Implement the final solution

Monitor the system

- Use the model to guide day-to-day operations

Monitor the system and update it regularly to meet organization's objectives

Conclusion

- OR **rationalizes complex** organizational management using scientific methodology
- OR involves **collaboration across multiple fields** to solve problems
- OR is about **understanding problems** and **proposing solutions** that **lead to cost savings and profits**

Conclusion

Real-world problem

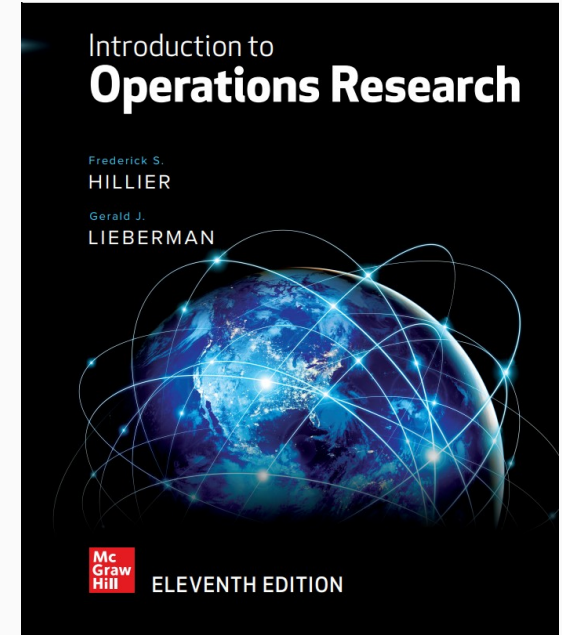
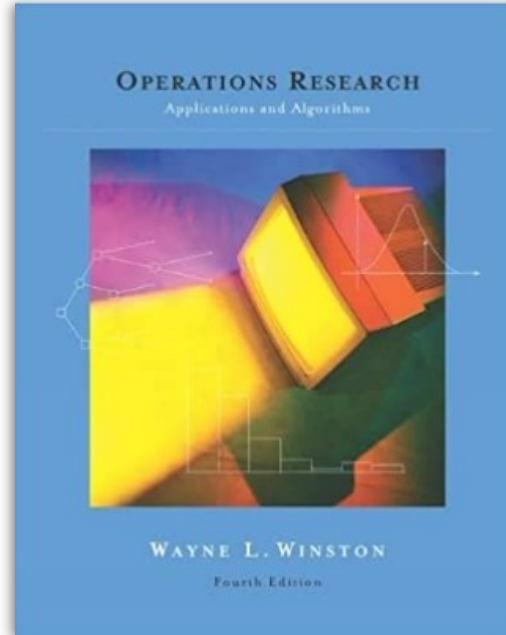
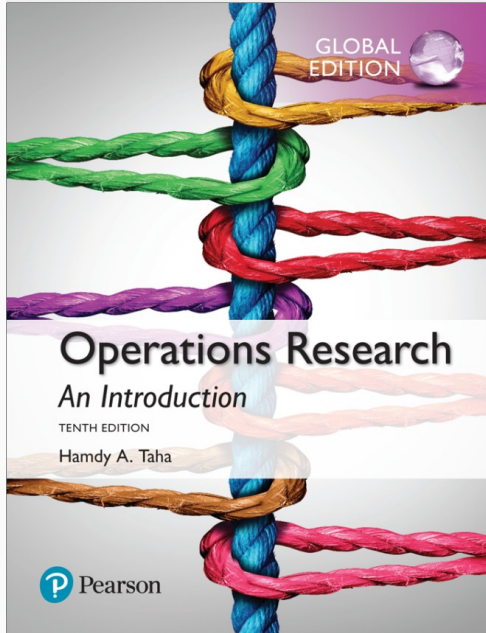
Abstract Model

model

Solve the

Improve the reality

Used Books for Lecture Preparation



Open discussion